PHS 2000B Problem Set 6: Instrumental Variables

Due Tuesday, April 2nd, 2026

We encourage students to help each other out on problem sets. While the final product that you turn in must be your own individual work that you have written up in isolation from your partners, you can still seek the help of your peers if you get stuck on a particular question. We request that you write the names of your collaborators below.

Collaborators:

Part 1: True, False, or Uncertain

Are the following statements True (T), False (F), or Uncertain (U). If true, state in a full sentence why this statement is true. If false, explain why and give an applied example of a situation where the statement would not be true. You should respond (U) if you believe the statement can be either T or F depending on what assumptions you make and you should give examples of both possibilities.

1) Assume that treatment A is offered at only one time point. If an instrument Z is randomized, it always satisfies the assumptions required to estimate causal treatment effects. (2 pts)

Solution

False. The exclusion restriction assumption requires that the instrument Z affects the outcome only through treatment A. While randomization of Z makes this assumption more likely to hold true, it may not always do so. For example, knowing one's treatment group assignment may impact behavior, choices, etc. that also affect outcome through pathways that do not involve the treatment. Moreover, we do not know whether Z is strong instrument.

2) If everyone in the population is a complier, then the LATE and the ATE are the same thing. (2 pts)

Solution

True. The LATE is the effect among those who are affected by the instrument in the expected manner (i.e. compliers). For example, if the instrument is randomization to treatment group, then those the compliers are those who take treatment when assigned treatment, and don't take treatment when not assigned treatment. ATE is the effect in the whole population. If everyone in the population is the complier, then the LATE applies to the whole population, and so is the same as the ATE. You can also think of the ATE as the weighted average of the effect of the exposure on the outcome among Always Takers, Never Takers, Compliers, and Defiers (see additional handout material for definitions). If everyone is a complier, you may think of the weights on all strata aside from compliers as being 0, which implies that the LATE = ATE.

3) For an instrument Z to be valid for exposure A, it should not predict the outcome Y.(2 pts)

Solution

False. Z should predict Y. The only restriction is that Z can affect Y **only through** A.

Part 2: IV assumptions

You have been asked to review a paper which proposes an instrument Z for an exposure A and outcome Y.

4) From the list below, select and explain which statements – if true – would make you doubt that Z is a good instrument for two-stage least squares estimation. (3 pts)

- The instrument Z's first stage F-statistic is close to 0.
- The instrument Z's first stage F-statistics is greater than 50.
- The instrument has a direct effect on the outcome Y
- The instrument has an indirect effect on the outcome Y only through A
- The instrument has an indirect effect on the outcome Y, but not through A (i.e., through some variable B that is unmeasured)
- The instrument may be correlated with the outcome Y through a common cause, C, that is unmeasured
- The instrument Z has also been used in other published research as an instrument for exposure B on outcome D. Outcome D is on the causal pathway to outcome Y, but D is not on the causal path to or from A

Solution

The first, third, fifth, sixth, and seventh statements would raise concerns about violations of the IV identification assumptions. The first two statements concern the relevance assumption. In practice, the rule of thumb is for an F-statistic to exceed 10, at a minimum. An F-statistic greater than 50 would suggest that Z is a strong instrument of A. The third statement is a violation of the exclusion restriction, which states that the instrument can have an indirect effect on the outcome Y only through A (which is itself stated in the fourth statement). The fifth statement, similarly, is a violation of the exclusion restriction. The sixth statement violates our assumption of independence, as an unmeasured common cause of both the outcome and the instrument is a back door path. Finally, the seventh statement suggests another violation of the exclusion assumption.

5) Your colleague is interested in identifying the average treatment effect (ATE) of A on Y using the instrument Z. She feels confident that there are no defiers in her population, but thinks it is plausible that there are always-takers and never-takers in her study. Under what assumptions can she still estimate an ATE? (2 pts)

Solution

Our colleague would need to make the assumption of no heterogeneity, which is equivalent to assuming homogeneity, or that the effect of A on Y is the same for every individual in the population. Note that our colleague wants the average treatment effect, not the LATE; merely assuming monotonicity (or no defiers) would only give the LATE.

6) List the key assumptions required for IV inference that CANNOT be verified. Choose one of the these assumptions and explain why the assumption cannot be empirically verified (i.e., why a statistical test or series of tests cannot show that the assumption is met). (2 pts)

Solution

There are four assumptions required for valid inference using IV: Relevance, Excludability, Independence, and Monotonicity. While the first can be tested, the other three cannot. We cannot statistically rule out that there is an unmeasured confounder of the instrument and outcome (independence) or that there is an indirect between them that does not flow through the exposure of interest (exclusion). Assumptions of homogeneity (or of the lesser assumption of monotonicity), are also difficult to test; while there are proposed methods that evaluate whether the assumptions may be falsifiable, there is no test that will verify the assumption that the treatment effect is the same across all individuals, or that there are no defiers in the population.

Part 3: Financial benefits of education

To estimate the income returns to education without confounding, the authors use the quarter of birth as an instrumental variable for schooling. The idea is that if you are born at the beginning of the year (i.e., the first quarter), then you start the school at an older age and therefore can drop out after completing less schooling than those who were born near the end of year. The authors use the two-stage least squares

regression to estimate the effects of schooling on wages. We analyze the data from the **1920** cohort from this study. The abstract from the paper is reproduced below:

> "We establish that season of birth is related to educational attainment because of school start age policy and compulsory school attendance laws. Individuals born in the beginning of the year start school at an older age, and can therefore drop out after completing less schooling than individuals born near the end of the year. Roughly 25 percent of potential dropouts remain in school because of compulsory schooling laws. We estimate the impact of compulsory schooling on earnings by using quarter of birth as an instrument for education. The instrumental variables estimate of the return to education is close to the ordinary least squares estimate, suggesting that there is little bias in conventional estimates."

For the remainder of this problem set, we will be delving into the study and developing an intuition for how IV estimation can be conducted in quasi-experimental studies like this one.

variable	description
`EDUC`	number of years in education
`LWKLWGE`	log weekly wage in 1980
`QOB`	quarter of birth
`first_quarter`	binary variable for whether the person was born in the first quarter

7) Why might an instrumental variable strategy be needed? That is, why can't the authors run a simple linear regression estimating the income returns to education? How would you expect the results to be biased? (2pt)

Solution

The authors would run into a number of validity issues if they conducted a simple linear regression. There are a number of observable and unobservable confounders (i.e. parental education, social networks, 'inate ability') that would bias their findings. Moreover, there is an issue of reverse causality, where we think that it is possible that a high paying job out out of high school may cause the individual to stop seeking higher education.

Students can argue for bias in either direction as long as the proposed confounding and its relationship to education and income are well articulated.

8) Estimate the first stage of the 2SLS estimator and interpret (2 point)

```
data <- read.csv("QOB20_phs2000b.csv")
head(data)
```

```
##   X.1  X EDUC LWKLYWGE QOB  YOB first_quarter
## 1   1 22   10     4.42   3 1920             0
## 2   2 30   16     6.07   4 1920             0
## 3   3 36   12     5.28   3 1920             0
## 4   4 37   12     5.84   2 1920             0
## 5   5 61    8     5.18   2 1920             0
## 6   6 64   16     5.89   2 1920             0
```

```
iv_first_stage <- lm(EDUC ~ first_quarter, data=data)
summary(iv_first_stage)
```

```
##
## Call:
## lm(formula = EDUC ~ first_quarter, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -11.282  -2.282   0.718   0.973   6.973
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    11.2819     0.0256  441.17  < 2e-16 ***
## first_quarter  -0.2546     0.0491   -5.18  2.2e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.36 on 23723 degrees of freedom
## Multiple R-squared:  0.00113,    Adjusted R-squared:  0.00109
## F-statistic: 26.9 on 1 and 23723 DF,  p-value: 2.19e-07
```

Solution

Being born in the first quarter causes a 0.2546 reduction in the number of years of education.

9) Evaluate whether first quarter birth is a strong instrument in this data and justify why or why not (1 point).

Solutions

As can be seen from the regression conducted in the previous question, this first stage has a F statistic of 26.9, which is higher than the traditional threshold of 10 for a strong instrument.

10) Can you identify some potential violations of the assumption of "excludability" in this setting? (2 point)

Answers may vary. Students will get full marks if they justify whether or not there is causal pathway from first quarter birth to income that is not mediated by years of education.

That being said, there is something called the Relative Age Effect (RAE), which is shown that those born in the first quarter perform better in sports and academics, which demonstrates that there is another causal pathway from quarter of birth and income.

11) Estimate the ITT and interpret (2 points)

```r
iv_itt <- lm(LWKLYWGE  ~ first_quarter, data=data)
summary(iv_itt)
```

```
##
## Call:
## lm(formula = LWKLYWGE ~ first_quarter, data = data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -5.158 -0.253  0.040  0.327  2.896
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.13828    0.00505 1018.38   <2e-16 ***
## first_quarter -0.02219    0.00969   -2.29    0.022 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.663 on 23723 degrees of freedom
## Multiple R-squared:  0.000221,   Adjusted R-squared:  0.000179
## F-statistic: 5.25 on 1 and 23723 DF,  p-value: 0.022
```

Solution

Being born in the first quarter causes a 0.02219 reduction in the average log-weekly wage in 1980.

Alternatively

Being born in the first quarter causes an approximate 2.2% reduction in the average weekly wage in 1980.

12) Estimate the LATE and its 95% confidence interval (*make sure to use a package that produces valid SEs like 'ivmodel' or 'feols'*). Interpret the LATE. (2 points)

```
library(ivmodel)
ivmodel(Y=data$LWKLYWGE, D=data$EDUC, Z=data$first_quarter)
```

```
##
## Call:
## ivmodel(Y = data$LWKLYWGE, D = data$EDUC, Z = data$first_quarter)
## sample size: 23725
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## First Stage Regression Result:
##
## F=26.9, df1=1, df2=23723, p-value is 2e-07
## R-squared=0.00113,   Adjusted R-squared=0.00109
## Residual standard error: 3.36 on 23724 degrees of freedom
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Coefficients of k-Class Estimators:
##
##              k Estimate Std. Error t value Pr(>|t|)
## OLS    0.00000  0.08081    0.00117   69.19   <2e-16 ***
## Fuller 0.99996  0.08694    0.03411    2.55    0.011 *
## TSLS   1.00000  0.08717    0.03474    2.51    0.012 *
## LIML   1.00000  0.08717    0.03474    2.51    0.012 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
##
## Alternative tests for the treatment effect under H_0: beta=0.
##
## Anderson-Rubin test (under F distribution):
## F=5.25, df1=1, df2=23723, p-value is 0.02
## 95 percent confidence interval:
##  [0.0146687797262122, 0.161791725203733]
##
## Conditional Likelihood Ratio test (under Normal approximation):
## Test Stat=5.25, p-value is 0.02
## 95 percent confidence interval:
##  [0.0146687760507622, 0.161791729097701]
```

Solution

$LATE = 0.08717$

$95\%CI = (0.08717 - 1.96 \times 0.03474, 0.08717 + 1.96 \times 0.03474)$ $95\%CI = (0.01908, 0.15526)$

Among those induced to receive less years of education due to being born in the first quarter, an additional year of education causes a 0.08717 increase in the average log-weekly wage in 1980.

Alternatively

Among those induced to receive less years of education due to being born in the first quarter, an additional year of education causes an approximate 8.7% increase in the average weekly wage in 1980.

13) Do you think the LATE estimated is homogenous or heterogenous. Justify your answer (2 sentences, 2 points).

Solution

To say that this LATE is homogenous (the same for compliers, always-takers, never-takers) would be the same as saying we have estimated the ATE of education on wages. However, given that the compliers in this case are those who only received less education because they reached the age minimum for dropping out earlier in high school. This would suggest that this a drastically different sub-group compared to the rest of the population. As such, it would not be reasonable to assume that the effect of education among this group would be the same as the rest of the population.